

Name - Saikat Sheet
Batch - 25SUB5936_Linux System Programming
ID - 68500

Task 11:

Inter-Process Communication (IPC) Using Shared Memory

- Implement an IPC mechanism using shared memory in C:
- Create a shared memory segment and attach it to the process's memory space.
- Write a string message to the shared memory in one process.
- Read and display the message from the shared memory in another process.

Code:

shm_writer

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <sys/stat.h>
#include <string.h>

int main() {
    const int seg_size = 1024;
    int shmid = shmget(IPC_PRIVATE, seg_size, IPC_CREAT | IPC_EXCL | 0666);
    if (shmid == -1) {
        perror("shmget");
        return 1;
    }
    printf("Writer: Shared memory ID: %d\n", shmid);

    char* shm = (char*)shmat(shmid, NULL, 0);
```

```

    if (shm == (char*)-1) {
        perror("shmat");
        return 1;
    }

    strcpy(shm, "Hello from shared memory!");

    shmdt(shm);
    printf("Writer: Wrote message and detached.\n");
    return 0;
}

```

shm_reader

```

#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>

int main() {
    int shmid;
    printf("Reader: Enter shmid from writer: ");
    scanf("%d", &shmid);

    char* shm = (char*)shmat(shmid, NULL, 0);
    if (shm == (char*)-1) {
        perror("shmat");
        return 1;
    }
    printf("Reader: Message: %s\n", shm);

    shmdt(shm);
    printf("Reader: Detached.\n");
    return 0;
}

```

Output:

```
May 9 19:59
student@student: ~/25SUB5936_U21/Assignments/Day30/Task11_Inter_Process_Communication
student@student: ~/25SUB5936_U21/Assignments/Day30/Task11_Inter_Process_Communication$ gcc shm_writer.c -o writer
student@student: ~/25SUB5936_U21/Assignments/Day30/Task11_Inter_Process_Communication$ ./writer
Writer: Shared memory ID: 4096055
Writer: Wrote message and detached.
student@student: ~/25SUB5936_U21/Assignments/Day30/Task11_Inter_Process_Communication$
```

```
May 9 20:00
student@student: ~/25SUB5936_U21/Assignments/Day30/Task11_Inter_Process_Communication
student@student: ~/25SUB5936_U21/Assignments/Day30/Task11_Inter_Process_Communication$ gcc shm_reader.c -o reader
student@student: ~/25SUB5936_U21/Assignments/Day30/Task11_Inter_Process_Communication$ ./reader
Reader: Enter shmid from writer: 4096055
Reader: Message: Hello from shared memory!
Reader: Detached.
student@student: ~/25SUB5936_U21/Assignments/Day30/Task11_Inter_Process_Communication$
```